Sreedhar Unnikrishnakurup

Senior Scientist, Structural Materials Department
Future Energy Acceleration and Translation (FEAT) Centre
Institute of Materials Research and Engineering (IMRE), A*STAR

2 Fusionopolis Way, 08-03 Innovis, Singapore 138634
sreedharun@a-star.edu.sg +65- 85897305
https://sites.google.com/site/sreedharunnikrishnakurup/home
https://www.linkedin.com/in/sreedhar-unnikrishnakurup-1a318866
https://orcid.org/0000-0001-6186-7562 DOB: 29 Aug 1984
H-index: 12: https://scholar.google.com/citations?user=rgeu6SAAAAAJ&hl=en



RESEARCH PROFILE

- Successfully led and secured **competitive research funding** as PI and Co-PI, including a multinational grant on curved composite inspection and a collaborative aerospace composite repair project under national R&D programs.
- Published **20 journals and 40 conference papers** in leading outlets focused on thermographic inspection of composites, demonstrating impact in the field of NDT of structural materials; additionally, filed **12 technology disclosures**, with one licensed to industry partners for practical implementation.
- Over **12 years** of post-PhD experience in advanced NDT, with expertise in Infrared Thermography (Pulsed, Lock-in, Induction, Vibro-, and Laser-based IRT), as well as hands-on work with Hyperspectral imaging, X-ray radiography, and THz imaging.
- Extensive experimental experience with sensor technologies, especially high-speed infrared cameras, and advanced NDT instrumentation for in-situ and lab-based inspection. **Level 2 certification** in Infrared Thermography.
- Proven ability to collaborate with multidisciplinary teams across academia and industry, with research experience spanning Europe and Asia.

RESEARCH SPECIALIZATIONS

- **Non-Destructive Testing & Evaluation (NDT&E):** Active Infrared Thermography (Pulsed, Lock-in, Induction, Vibro-, and Laser-Assisted IRT), Hyperspectral & Multispectral Imaging, Terahertz Time-Domain Spectroscopy (THz-TDS), Shearography, X-ray Radiography & Computed Tomography (CT).
- **Structural Health Monitoring (SHM) & Advanced Materials:** Polymer Matrix Composites (CFRP/GFRP), Aerospace Composite Repair & Honeycomb Structures, Lightning Strike & Impact Damage Assessment, Thermal Barrier Coatings, In-situ Monitoring of Laser Powder Bed Fusion (LPBF) Additive Manufacturing.
- **Computational Modeling, Signal Processing & AI:** Multiphysics Finite Element Modeling (COMSOL), Inverse Heat Conduction & Anisotropic Heat Diffusion, Thermal Signal Processing, Machine Learning & Deep Learning for Automated Defect Recognition (ADR).

WORK EXPERIENCE

Senior Scientist

Structural Materials Department, Institute of Materials Research and Engineering (IMRE), A*STAR

Future Energy Acceleration and Translation (FEAT) Centre, IMRE, A*STAR (2026–Ongoing)

Apr 2024–Ongoing Singapore

- Future Energy Acceleration and Translation (FEAT) Centre (**AMOFES** & **STRIDE** Projects): Advanced non-destructive evaluation and pulsed laser thermography characterization of extreme-environment and fusion energy materials, including interfacial thermal resistance assessment of tungsten coatings on boron carbide substrates [C2].
- Singapore Aerospace Programme (SAP) (Co-PI): UD-CFRP cross-ply laminate characterization using Induction Thermography for aerospace composite repair [J3, J6, C6].
- Material Informed Differential Generative AI (MIDGAI) for Multi-functional Composites (Project Member): Development of high-strength and high-conductivity composites using generative AI.
- Lead Researcher on Advanced NDT Projects: Multimodal NDE using IRT, Ultrasonic, and Shearography of lightning strike damage in painted CFRP laminates with lightning strike protection [J1]; sweep-excited vibrothermography and machine learning for low-energy impact damage in carbon composites [C3]; infrared thermography diagnostics for reverse-bias stability in perovskite solar modules [J2]; water ingress detection in honeycomb sandwich aircraft structures; evaluation of cold-sprayed coating porosity [J5]; sub-coating crack detection in metallic components using induction IRT; and physics-informed AI/deep learning for automated defect detection in IRT [J4].

Scientist

Structural Materials Department, Institute of Materials Research and Engineering (IMRE), A*STAR

Nov 2021–Mar 2024 — Singapore

- Project Polymer Matrix Composites Programme (PMCP) (Work Package Lead): Managed work package involving in-line IRT monitoring of prepreg manufacturing, characterization of impact and manufacturing damages in CFRP [J7, C4, C12], inspection of filament-wound GFRP pipes, and impact damage characterization using Induction IRT.
- Principal Investigator (PI) on Multinational Collaborative Grant: Developed quantitative active thermography and optical profile correction methods for inspecting complex curved aerospace composite structures [C5, C7, C8].

Visiting Researcher

Mechanical Design and Machine Construction (IIND-03/A) NDT Research Group, Politecnico di Bari

November 2025 - February 2026 — Bari, Italy

- Thermal Signature Analysis for Defect Detection in Laser-Based Additive Manufacturing [C1]
- Development of Machine Learning Algorithms for the inline monitoring of LPBF process

Senior Project Officer

Centre for Non-Destructive Evaluation (CNDE), IIT Madras

May 2018 - Sep 2021 — Chennai, India

- Application of Nondestructive IRT testing for thickness inspection of thermal barrier coatings and validation using THz-TDS and Eddy current testing [J12].
- Composite inspection using induction heating IRT. [J9]
- Crack inspection in steel billets at elevated temperatures using Laser assisted IRT [J14].
- Development of artificial intelligence-assisted advanced X-ray radiography imaging and automatic defect recognition of critical welds and high-energy materials [J10]
- Numerical investigation of anisotropic heat diffusion in polycrystalline and composite microstructures [J13].

Cohort Member BA3

Entrepreneur First

February 2020 - May 2020 — Bangalore, India

- Explored several interesting technological solutions, exploring the market, validating the technology, speaking to customers, and researching prototypes
- The ideas that we worked on is, detect and report quality-compromised and counterfeit drugs using a mobile cloud-based cost-effective hyper-spectral Imaging based AI device

Postdoctoral Research Scientist

NDT group 8.7, Federal Institute for Material Research and Testing (BAM)

January 2016 - June 2017 — Berlin, Germany

- Project VITCEA (Validated Inspection Techniques for Composites in Energy Applications)
- Optimize and validate active thermography as full-field, fast and non-contact NDE technique for quantitative testing of FRP structures [J16]
- Laser Thermography for surface crack detection at high temperature [C23]

Visiting Research Scientist

Electromagnetic NDT Research Group, West Pomeranian University (ZUT)

January 2015 - March 2015 | Szczecin, Poland

- Scientific Exchange as part of FP7 Project: Health Monitoring of Offshore Wind Farms (HEMOW)
- Terahertz imaging and time domain analysis for material defect identification in Composite Wind turbine blades [C33]
- Active Infrared Thermography for defect identification in composite materials - Numerical investigations of heat sink effects in the infrared inspecton of composites [C29, C34]

Institute Postdoctoral Fellow

Centre for Non-Destructive Evaluation (CNDE), IIT Madras

September 2014 - December 2015 | Chennai, India

- Numerical modelling of microstructural heat diffusion and ultrasonic wave propagation in Polycrystalline Materials [C25, C27, C28]
- CMT welding process monitoring using IR thermography [J18, C21, C24]
- In-line laser thermography for Online Monitoring of steel billets to predict the surface cracks formed during the non-uniform cooling [C26, C30]

Research Associate

Centre for Non-Destructive Evaluation (CNDE), IIT Madras

April 2007 - December 2008 | Chennai, India

- Development of online weld quality monitoring for AA2219 liquid propellant tanks using Infrared Thermal Imaging and X-ray radiography [J20]
- Developed a finite element model for the prediction of temperature profiles during welding

EDUCATION

Ph.D. in Mechanics & Civil Engineering

Department of Mechanics and Civil Engineering (LMGC), University of Montpellier 2 (UM2), France

2011 - 2014 | Montpellier, France

- Dissertation Topic: Weld pool shape identification using Multi-physics modelling and Experiments [J17, C35-C39]

M.S. in Mechanical Engineering

Department of Mechanical Engineering, Indian Institute of Technology Madras (IIT Madras), India

2008 - 2010 | Chennai, India

- Dissertation Topic: Online weld quality monitoring using Infrared Thermal Imaging [J19, J20, C32, C40]

Bachelors in Mechanical Engineering

Department of Mechanical Engineering, Institution of Engineers (India), India

2002 - 2006 | Kottayam, India

HONORS, AWARDS AND COMPETITIVE GRANTS

- Conference Co-Chair, Asian Quantitative Infrared Thermography Conference (QIRT-Asia 2027), Singapore — 2027
- Invited Speaker Recognition, NDE 2026 Conference, Indian Society for Non-Destructive Testing (ISNT) — 2026
- Visiting Researcher Fellowship, NDT Research Group, Politecnico di Bari, Italy — 2025–2026
- Technology Commercialization & Licensing Recognition (1 TD Licensed, 12 TDs Filed), IMRE, A*STAR — 2021–2026
- Steering Committee Member, Quantitative InfraRed Thermography (QIRT & QIRT-Asia) — 2023–Present
- Co-Principal Investigator, Singapore Aerospace Programme (SAP) Research Grant, A*STAR — 2022–2026
- Principal Investigator, Multinational Grant on Curved Aerospace Composite Inspection, A*STAR — 2022–2025
- Work Package Lead, Polymer Matrix Composites Programme (PMCP), A*STAR, Singapore — 2021–2025

INVITED PRESENTATIONS

- **Invited Speaker**, NDE 2026 Conference & Exhibition, Organized by Indian Society for Non-Destructive Testing (ISNT), Hyderabad, India, November 14–17, 2026.
- **Invited Research Seminar**, Mechanical Design and Machine Construction (IIND-03/A) NDT Research Group, Department of Mechanics, Mathematics and Management (DMMM), Politecnico di Bari, Talk: *Thermal Signature Analysis and Machine Learning for In-line Monitoring of Additive Manufacturing*, Bari, Italy, 2025–2026.
- **Invited Tutorial Speaker**, Preconference Tutorials, NDE 2019 Organized by Indian Society for Non-Destructive Testing (ISNT), Talk: *Applications of Thermal Imaging in Aerospace Industry*, Bengaluru, India, December 3, 2019.
- **Invited Speaker**, Workshop on Advanced NDE Techniques and Applications organized by TATA STEEL, Talk: *Advanced Thermography Techniques*, TATA Nagar, Jamshedpur, India, August 17, 2015.
- **Invited Speaker**, Polish Society for Theoretical and Applied Electrical Engineering Section in Szczecin (PTETiS), Talk: *Research and Development in the Field of NDE @ CNDE*, West Pomeranian University of Technology, Szczecin, Poland, March 12, 2015.
- **Invited Speaker**, Polish Society for Nondestructive Testing and Technical Diagnostics Section in Szczecin (PTBNiDT), Talk: *Research and Applications of IR Thermography @ CNDE*, Szczecin, Poland, February 26, 2015.

CERTIFICATIONS AND MEMBERSHIPS

- Infrared Thermography Level 2 Certification. Issued on October 2026
- Project Management Fundamentals by EPM. Issued on October 2022
- Infrared Thermography Level 1 by Infrared Training Center. Issued on May 2022
- Deep Learning, a 5-course specialization by deeplearning.ai on Coursera. Specialization Certificate earned on September 23, 2018
- Technical Committee Member, ISO/TC 135/SC 8 – Thermographic Testing
- Member of ASTM International
- Quantitative InfraRed Thermography (QIRT) Steering Committee Member
- Asian Quantitative Infrared Thermography (QIRT-Asia) Steering Committee Member
- Member of Non-Destructive Testing Society Singapore (NDTSS)
- Member of Indian Society for Non-Destructive Testing (ISNT)
- Member of Institution of Engineers (India), and Chartered Engineer Certification

MENTORING

During postdoctoral tenure at IIT Madras: 4 Ph.D. Students and 2 Master student

- Shivaprasad S (Ph.D.) - Voronoi Cell based FEM simulations for wave propagation in solids [C25, C27]
- Renil Thomas K (Ph.D.) - Induction thermography inspection of CFRP and CMT weld monitoring [J3, J6, J9, J18]
- Nithin P V (Ph.D.) - Laser thermography for crack inspection in steel samples at elevated temperature [J14, J15]
- Surendra Gupta (Ph.D.) - Emissivity characterization of nano photonic emitters [J11]
- Anurag Saini (MTech) - Numerical investigation of Heat diffusion in particle reinforced composite [J13]
- Dhruv Gamdha (MTech) - Ray tracing model for X-ray simulation and AI based automatic defect Recognition [J10]

PUBLICATION

Journal Articles

J1. **Sreedhar U**, Santhakumar Sampath, Zi Wen Tham, Siok Wei Tay, Jonathan Zheng, Vinod Kumar, Warintorn Thitsartarn, Lei Zhang, and Andrew Ngo. Multimodal non-destructive evaluation of lightning strike damage in painted CFRP laminates: An infrared thermography-based damage index approach with cross-validation. *NDT & E International*, page 103817, 2026. link

J2. Xi Wang, Ran Luo, Nengxu Li, Jia Li, Tao Wang, Julian A. Steele, Eduardo Solano, Zihao Zhu, Xinxing Yin, Jinxi Chen, **Sreedhar U**, Andrew Ngo, Yuduan Wang, Jingcong Hu, Zhenrong Jia, Zhuojie Shi, Xinyi Du, Xiuxiu Niu, and Yi Hou. Molecular-templated pre-assembly of self-assembled monolayer for perovskite solar cells and modules with improved reverse-bias stability. *Nature Energy*, pages 1–12, 2026. link

J3. **Sreedhar U**, Renil K T Kidangan, CV Krishnamurthy, K Balasubramaniam, and Andrew Ngo. Assessment of fiber orientation and order in carbon fiber reinforced polymer composites using induction thermography and radon transform analysis. *Composites Part B: Engineering*, 296:112224, 2025. URL https://bit.ly/4c1dzsB

J4. Leith Bounenni, Mohamed Arbane, Clemente Ibarra-Castanedo, Yacine Yaddaden, **Sreedhar U**, Andrew N C Yong, and Xavier Maldague. Advanced defect detection on curved aeronautical surfaces through infrared imaging and deep learning. *NDT*, 2(4):519–531, 2024. URL https://bit.ly/4hi8sW8

J5. **Sreedhar U**, Zheng Zhang, D H Leng Seng, Zhi-qian Zhang, Jisheng Pan, Vinod Kumar, Jonathan Zheng, Carlos Manzano, and Andrew Ngo. Exploring thermal dynamics and porosity of cold-sprayed ti-6al-4v coatings on al6061-t6 substrates: A pulsed thermography and numerical modeling approach. *International Journal of Thermal Sciences*, 196:108732, 2024. URL https://bit.ly/4kDFovc

J6. Renil K T, **Sreedhar U**, CV Krishnamurthy, and Krishnan Balasubramaniam. Uncovering the hidden structure: A study on the feasibility of induction thermography for fiber orientation analysis in cfrp composites using 2d-fft. *Composites Part B: Engineering*, 269:111107, 2024. URL https://bit.ly/4igfbB6

J7. Renil K T, **Sreedhar U**, CV Krishnamurthy, and K Balasubramaniam. Induction thermography for unidirectional cfrp composites: A novel inspection approach through global current path integration. *Composite Structures*, 327:117678, 2024. URL https://bit.ly/41wBBXL

J8. Mercy L, **Sreedhar U**, J Abhinandan, M K Patra, and K Balasubramaniam. Material characterization and thickness measurement of iron particle reinforced polyurethane multi-layer coating for aircraft stealth applications using thz-time domain spectroscopy. *Journal of Infrared, Millimeter, and Terahertz Waves*, 2022. URL https://bit.ly/3vQhipj

J9. Renil Thomas Kidangan, Sreedhar Unnikrishnakurup, Chitti Venkata Krishnamurthy, and Krishnan Balasubramaniam. The influence of interlaminar microstructure on the induction heating patterns of cfrp laminates. *Materials Today Communications*, 33:104338, 2022. URL https://bit.ly/41Yc89B

J10. D Gamdha, **Sreedhar U**, and K Balasubramaniam. Automated defect recognition on x-ray radiographs of solid propellant using deep learning based on convolutional neural networks. *Journal of Nondestructive Evaluation*, 40(18), 2021. URL https://bit.ly/3aLxA7k

J11. Surendra G, Ameen E, **Sreedhar U**, K Balasubramaniam, A Veeraragavan, and B Pesala. Spectral filtering of sub-bandgap radiation using all-dielectric gratings for thermophotovoltaic applications. *Journal of Photonics for Energy*, 11(1):015501, 2021. URL https://bit.ly/3paK9hv

J12. **Sreedhar U**, J Dash, S Ray, B Pesala, and K Balasubramaniam. Nondestructive evaluation of thermal barrier coating thickness degradation using pulsed IR thermography and THz TDS measurements: A comparative study. *NDT & E*, 116: 102367, 2020. URL https://bit.ly/2GJUUHw

J13. A Saini, **Sreedhar U**, C V Krishnamurthy, T Sundararajan, and K Balasubramaniam. Numerical study using finite element method for heat conduction on heterogeneous material with varying volume fraction, shape and size of filler in matrix. *International Journal of Thermal Sciences*, 159:106545, 2020. URL https://bit.ly/31qVIYg

J14. P V Nithin, R T Kidangan, **Sreedhar U**, P Myrach, M Ziegler, and K Balasubramaniam. Laser line scanning thermography for surface breaking crack detection: modeling and experimental study. *Infrared Physics & Technology*, 104:103141, 2020. URL https://bit.ly/2YvQJ7h

J15. P V Nithin, R T Kidangan, **Sreedhar U**, P Rajagopal, K V Phani Prabhakar, G Padmanabham, and K Balasubramaniam. Numerical model and experimental validation for the on-line monitoring of cold metal transfer joining of aluminium to galvanized steel. *International Journal of Advanced Manufacturing Technology*, 104:4365–4375, 2019. URL https://bit.ly/2AnbYQF

J16. C Maierhofer, R Krankenhagen, M Roellig, **Sreedhar U**, C Monte, A Adibekyan, B Gutschwager, L Knazowicka, A Blahut, and M Gower. Influence of thermal and optical material properties on the characterization of defects in fiber reinforced composites with active thermography methods. *tm-Technisches Messen*, 85(1):13–27, 2018. URL https://bit.ly/3hswSOI

J17. **Sreedhar U**, S Rouquette, F Soulié, and G Fras. Estimation of heat flux parameters during static gas tungsten arc welding spot under argon shielding. *International Journal of Thermal Sciences*, 114:205–212, 2017. URL https://bit.ly/3d10rnl

J18. R T Kidangan, **Sreedhar U**, PV Nithin, K Balasubramaniam, P Rajagopal, KV Phani Prabhakar, G Padmanabham, F Riedel, and M Puschmann. Online monitoring of cold metal transfer (CMT) process using infrared thermography. *Quantitative InfraRed Thermography Journal*, 14(1):68–78, 2017. URL https://bit.ly/2zyzWrI

J19. **Sreedhar U**, CV Krishnamurthy, and K Balasubramaniam. Monitoring tig welding using infrared thermography–simulations and experiments. *Przegląd Elektrotechniczny*, 92(4):6–9, 2016. URL https://bit.ly/2UGtp5I

J20. **Sreedhar U**, CV Krishnamurthy, K Balasubramaniam, VD Raghupathy, and S Ravisankar. Automatic defect identification using thermal image analysis for online weld quality monitoring. *Journal of Materials Processing Technology*, 212(7): 1557–1566, 2012. URL https://bit.ly/3d5JMiA

Conference Presentations

C1. E. D'Accardi, **Sreedhar U**, D. Palumbo, A. Ngo, and U. Galietti. Multi-layer thermal feature analysis for defect detection and characterization in AlSi10Mg parts produced by laser powder bed fusion. In *The 18th International Conference on Quantitative InfraRed Thermography*, July 2026

C2. **Sreedhar U**, Vinod Kumar, Jonathan Zheng, Nancy Wong, Shi Jie Wang, and Andrew Ngo. Pulsed laser thermography for interfacial thermal resistance characterisation of tungsten coatings on boron carbide substrates. In *The 18th International Conference on Quantitative InfraRed Thermography*, July 2026

C3. **Sreedhar U**, Keefe Wee, Santhakumar Sampath, and Andrew Ngo. Enhancing low-energy impact damage detection in carbon composites using sweep-excited vibrothermography and machine learning. In *The 18th International Conference on Quantitative InfraRed Thermography*, July 2026

C4. **Sreedhar U**, M. Sharma, A. Feng, J. Zheng, V. Kumar, C. Manzano, Z. Wong, and A. Ngo. Lock-in thermography and machine learning-based defect detection in carbon fiber reinforced thermoplastics. In *Singapore International NDT Conference & Exhibition*. e-Journal of Nondestructive Testing, February 2025. URL https://www.ndt.net/?id=30843. Held on 27 - 28 February 2025

C5. J Zheng, V Kumar, **Sreedhar U**, C Manzano, S Ebrahimkhani, VV Tuan, LH Anh, N-M Cheung, and ACY Ngo. Multispectral and hyperspectral infrared imaging for plastic waste sorting and recycling. In *Thermosense: Thermal Infrared Applications XLVI*, volume 13047, pages 53–62, 2024. URL https://doi.org/10.1117/12.3013731

C6. **Sreedhar U**, Renil Thomas Kidangan, CV Krishnamurthy, and Krishnan Balasubramaniam. Characterizing fiber orientations and layer stacking sequences in cfrp laminates through induction heating patterns: a nondestructive approach. In *Proceedings of 4th Asian Quantitative InfraRed Thermography Conference Characterizing*, 2023. URL https://doi.org/10.21611/qirt.2023.17

C7. S.O.A. Gnessougou, S. Unnikrishnakurup, A. Ngo Chun Yong, P. Servais, C. Ibarra-Castanedo, and X. P. V. Maldague. Curved cfrp specimen: Thermographic inspection and simulation. In *Proceedings of 4th Asian Quantitative InfraRed Thermography Conference Characterizing*, 2023. URL https://doi.org/10.21611/qirt.2023.22

C8. **Sreedhar U**, Vinod Kumar, Jonathan Zheng, Carlos Manzano, and Andrew Ngo. Exploring ambient influences on infrared thermography: A study on concrete delam-ination detection and solar loading. In *Proceedings of 4th Asian Quantitative InfraRed Thermography Conference Characterizing*, 2023. URL https://doi.org/10.21611/qirt.2023.21

C9. Vinod Kumara, Carlos Manzano, **Sreedhar U**, Jonathan Zheng, and Andrew Ngob. Dynamic inspection of surface-breaking defects using induction thermography. In *Singapore International Non-destructive Testing Conference and Exhibition (SINCE2022)*, Singapore, November 2022. URL https://doi.org/10.58286/27534

C10. Jonathan Zheng, Carlos Manzano, **Sreedhar U**, Vinod Kumar, and Andrew Ngo. Using hyperspectral imaging as a non-destructive method to discern artworks. In *Singapore International Non-destructive Testing Conference and Exhibition (SINCE2022)*, Singapore, November 2022. URL https://doi.org/10.58286/27551

C11. **Sreedhar U**, Jonathan Zheng, Vinod Kumar, Carlos Manzano, and Andrew Ngo. Crack detection in metallic components with varying surface characteristics using laser spot scanning thermography. In *Singapore International Non-destructive Testing Conference and Exhibition (SINCE2022)*, Singapore, November 2022. URL https://doi.org/10.58286/27536

C12. **Sreedhar U**, Zheng J, Manzano C, and Ngo C, Y. Numerical and experimental investigations of pulsed infrared thermography for damage detection in honeycomb sandwich composites. In *16th Quantitative Infrared Thermography Conference*, Paris, France, July 2022. URL https://doi.org/10.21611/qirt.2022.2025

C13. J Dash, **Sreedhar U**, B Pesala, and K Balasubramaniam. Material parameters and thickness determination of thermal barrier coatings using terahertz time domain spectroscopy. In *Virtual Conference and Exhibition on Nondestructive Evaluation (NDE)*, Virtual, December 2020. URL https://www.ndt.net/?id=26534

C14. Surendra G, Ameen E, A Veeraraghavan, **Sreedhar U**, K Balasubramaniam, and B Pesala. High contrast grating based broadband thermal filters for thermophotovoltaic applications. In *SPIE. Photonic West OPTO*, San Francisco, US, February 2020

C15. D Gamdha, S Dwarakanath, **Sreedhar U**, and K Balasubramaniam. Simulation assisted annotated xray image data generation for deep learning applications. In *National Seminar and Exhibition on Non-Destructive Evaluation NDE*, Bangalore, India, December 2019

C16. **Sreedhar U**, J Dash, B Pesala, and K Balasubramaniam. Non-destructive evaluation of thermal barrier coating thickness degradation using pulse phase thermography and THz-TDS. In *11th International Symposium on NDT in Aerospace*, Paris-Saclay, France, November 2019. URL https://bit.ly/2ZDVsEn

C17. S Krishna, **Sreedhar U**, and K Balasubramaniam. Evaluation of thermal barrier coating thickness using pulsed thermography experiments and numerical simulations. In *National Seminar and Exhibition on Non-Destructive Evaluation NDE*, Mumbai, India, December 2018

C18. P V Nithin, R T Kidangan, **Sreedhar U**, Krishnamurthy C, V, M Zeigler, P Myrach, and K Balasubramaniam. Numerical study of laser line thermography for crack detection at elevated temperature. In *14th Quantitative Infrared Thermography Conference*, Berlin, Germany, June 2018. URL https://bit.ly/2VLk0u8

C19. C Maierhofer, R Krankenhagen, M Röllig, **Sreedhar U**, C Monte, A Adibekyan, B Gutschwager, L Kanzowicka, A Blahut, Gower M R L, M Lodeiro, Baker G, and Aktas A. Einfluss thermischer und optischer materialeigenschaften auf die charakterisierung von fehlstellen in faserverbund-werkstoffenmit aktiven thermografieverfahren. In *Temperature*, Berlin, Germany, May 2017. URL https://bit.ly/2C7nJuS

C20. S Krishna, P V Nithin, R T Kidangan, **Sreedhar U**, M Zeigler, P Myrach, K Balasubramaniam, and Biju P. Raw data-based image processing algorithm for fast detection of surface breaking cracks. In *43rd Annual Review of Progress in Quantitative Nondestructive Evaluation, AIP Conference Proceedings*, Atlanta, Georgia, US, July 2017. URL https://bit.ly/3e2aEQW

C21. R T Kidangan, **Sreedhar U**, K Balasubramaniam, L Narayanan, and G Phanikumar. Dissimilar metal joint quality measurement using infrared thermography: Experimental and numerical approach for the application to cmt welding. In *Quantitative InfraRed Thermography*, Gdańsk, Poland, July 2016. URL https://bit.ly/3e1f8Y7

C22. P V Nithin, S Krishna, R T Kidangan, **Sreedhar U**, C V Krishnamurthy, M Zeigler, P Myrach, and K Balasubramaniam. In-line laser thermography for crack detection at elevated temperature: A numerical modeling study. In *Quantitative InfraRed Thermography*, Gdańsk, Poland, July 2016. URL https://bit.ly/2ZOCBV2

C23. B Polomski, P Myrach, E Le Claire, **Sreedhar U**, P V Nithin, K Balasubramaniam, and M Ziegler. Thermographic crack detection in hot steel surface. In *19th World Conference on Non-destructive Testing (WCNDT)*, Munich, Germany, June 2016. URL https://bit.ly/3grMvER

C24. R T Kidangan, **Sreedhar U**, P V Nithin, K Balasubramaniam, P Rajagopal, V Phani Prabhakar, K, and G Padmanabham. Application of infrared thermography technique for the monitoring of cold metal transfer (cmt) joining of aluminium to galvanized steel. In *19th World Conference on Non-destructive Testing (WCNDT)*, Munich, Germany, June 2016. URL https://bit.ly/2NYJqQp

C25. Shivaprasad S, **Sreedhar U**, S Natarajan, C V Krishnamurthy, and K Balasubramaniam. 3d elastic wave propagation and heat diffusion studies using polycrystalline material models. In *National Seminar and Exhibition on Non-Destructive Evaluation NDE*, Hyderabad, India, November 2015. URL https://bit.ly/2NYasaM

C26. **Sreedhar U**, P V Nithin, C V Krishnamurthy, M Zeigler, P Myrach, and K Balasubramaniam. The effect of surface breaking crack orientation in detection capability: A laser thermography numerical modeling approach. In *National Seminar and Exhibition on Non-Destructive Evaluation NDE*, Hyderabad, India, November 2015

C27. Shivaprasad S, **Sreedhar U**, C V Krishnamurthy, and K Balasubramaniam. Elastic wave propagation and heat diffusion studies in polycrystalline material. In *Comsol Conference*, Pune, India, October 2015. URL https://bit.ly/2BE2YqQ

C28. **Sreedhar U**, C V Krishnamurthy, and K Balasubramaniam. Heat diffusion in polycrystalline materials- a microstructure-based material model. In *Quantitative InfraRed Thermography Conference QIRT-Asia*, Mahabalipuram, India, July 2015. URL https://bit.ly/2BEf0R1

C29. B Szymznik, **Sreedhar U**, T Chady, and K Balasubramaniam. Numerical analysis of heat sink effect in the infrared inspection of composites. In *Quantitative InfraRed Thermography Conference QIRT-Asia*, Mahabalipuram, India, July 2015. URL https://bit.ly/3dZVz2m

C30. **Sreedhar U**, P V Nithinn, C V Krishnamurthy, M Zeigler, P Myrach, and K Balasubramaniam. In-line laser thermography for crack detection: A numerical approach. In *Quantitative InfraRed Thermography Conference QIRT-Asia*, Mahabalipuram, India, July 2015. URL https://bit.ly/3iwPe1F

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Innovation and Technology Transfer

- Filed **12 Technology Disclosures** at Institute of Materials Research and Engineering (IMRE), A*STAR between 2021–2026 in the areas of Infrared Thermography, Composite Inspection, and Laser-Based NDT.
- **1 TD licensed** to industry partner for use in composite inspection workflows in bicycle repair.